

Navodit Maheshwari

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Education

Masters of Science Natural Language Processing, UC Santa Cruz Sept 2025 – Dec 2026

- GPA: 4.0/4.0
- **Coursework:** Deep Learning for Natural Language Processing, Data Science and Machine Learning Fundamentals, Natural Language Processing I,II, Conversational Agents

Bachelors of Science in Computer Science, UC Santa Cruz Sept 2022 – June 2025

- GPA: 3.5/4.0
- **Coursework:** Artificial Intelligence, Intro to Natural Language Processing, Machine Learning, Principles of Computer Systems Design, Computer Architecture, Introduction to Probability Theory

Skills

Programming Languages: Python, R, SQL, Bash, C/C++ , Java

Machine Learning Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, Pandas, NumPy

Data Visualization: Matplotlib, Seaborn, Plotly, Databricks, Spark, Ray

Tools & Platforms: Git, Docker, MLflow, Weights & Biases, Jupyter Notebook, UNIX/Linux, macOS, Windows, AWS/GCP/Azure

Experience

Graduate Researcher – LLM Steering & Alignment, Lab of Chenguang Wang Feb 2026 – Present

- Identified activation entanglement in single-layer DIM steering vectors, showing consistent out-of-distribution degradation across three LLM families.
- Developed a multi-layer steering approach that redistributes high-magnitude activations across transformer layers, reducing unintended side effects.
- Used GemmaScope and SAELens to analyze representation quality through reconstruction error, sparsity (L0), and entropy-based metrics.
- Evaluated steering effectiveness and safety across 17 datasets on Gemma 3 4B, Llama 3.1 8B, and Qwen 3 4B.

Tutor for Engineering Principles of Electronics, Baskin Engineering – Santa Cruz, CA Jan 2026 – Present

- Supported over 200 students in a cross-disciplinary course covering sensors, analog/digital electronics, and communication systems.
- Provided detailed, rubric-aligned feedback on assignments to help students improve quickly and understand core concepts.
- Answered student questions on course material, assignments, and logistics with clear and timely communication.

Projects

X-Ray Fracture Detection - GitHub Mar 2026

- Fine-tuned CLIP, BLIP, and BiomedCLIP on 60K MIMIC-CXR images to generate structured fracture reports.
- Used hard negative mining to handle 98% normal scans and improve contrastive learning.
- Benchmarked against ClinicalBLIP using METEOR; achieved 0.276 with BiomedCLIP (3.4× over zero-shot baseline).

AI Fitness Agent - GitHub Mar 2026

- Built a multi-agent fitness system using GPT-OSS 20B, Qwen2.5-7B, and Gemma3-27B for personalized workouts.
- Integrated Google Calendar and Health APIs to align workouts with schedules and biometric data.
- Implemented RAG to ground recommendations in user history, habits, and available equipment.
- Developed a vision agent to detect gym equipment from images and map to muscle groups.